



Effects of Mach number and 3D instabilities on boundary-layer transition over gaps

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IMPERIAL



Motivation

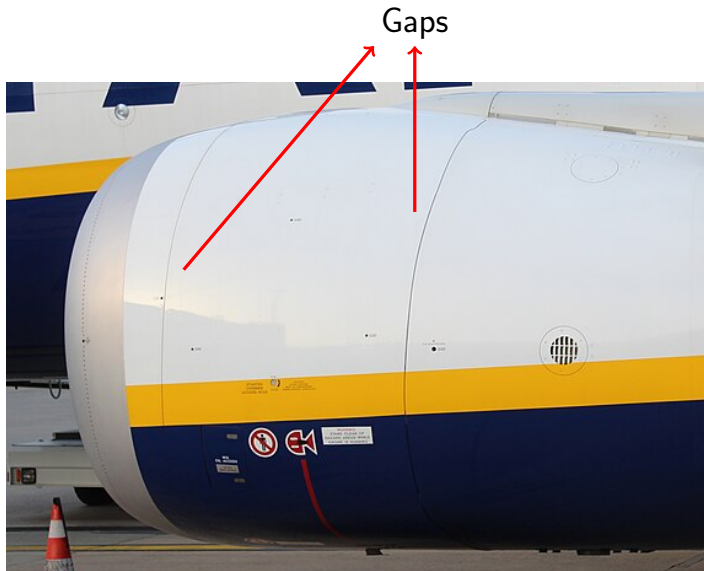


Figure: Engine nacelle of a Boeing 737

First questions

- Small disturbances in the boundary layer initially grow as $\propto e^{n(x)}$.
- We aim to understand how much a gap of width w and depth d can overamplify these disturbances.
- For each gap of width w and depth d , we define

$$\Delta n_{\text{gap}} := n_{\text{gap}} - n_{\text{flat}}$$

Experimental observations

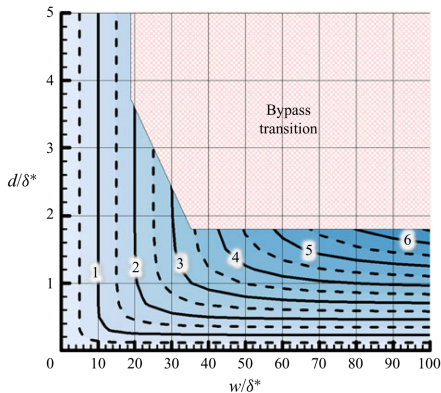
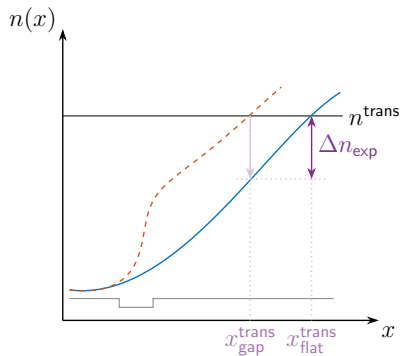


Figure: Δn_{gap} contours in the w - d parameter space, from experiments by Crouch et al. (2022)[†].

[†] Crouch JD, Kosorygin VS, Sutanto MI, Miller GD. Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability. *Flow*. 2022;2:E8.

Numerical setup

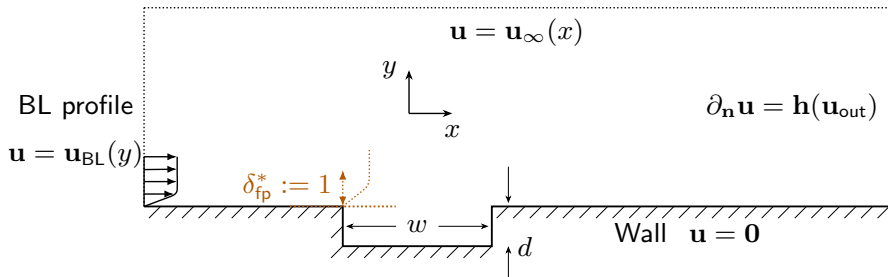


Figure: Domain setup for the steady-state finder

Aims:

- Study the stability of the system as a function of the depth d and width w of the gap. Reynolds number is fixed to

$$\text{Re} = \frac{u_{\infty} \delta_{fp}^*}{\nu} = 1000.$$

- Understand the influence of the Mach number.
- To what extent are 3D effects important?

Mesh and computational price

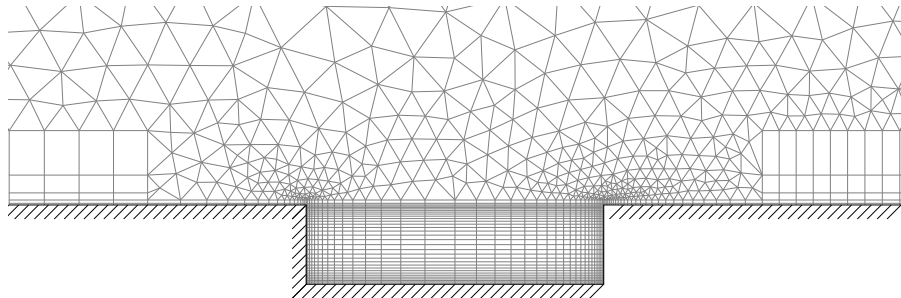


Figure: Mesh of a gapped domain

- Num. elements: $\sim 8\,000$, $p = (4, 5) \Rightarrow \sim 90\,000$ DOFs
- Time to run 24 000 time units: ~ 13 h on 32 cores.

2d incompressible limit set classification

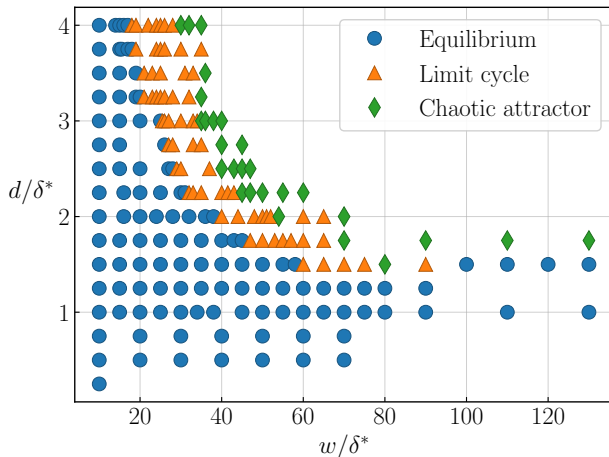
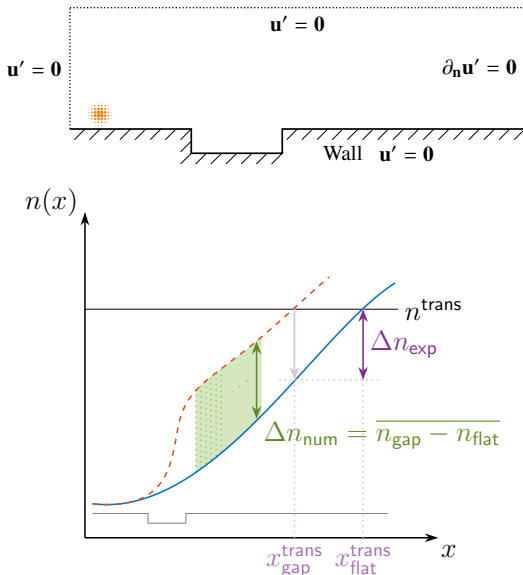


Figure: Limit set classification over the parameter space $10 \leq w/\delta^* \leq 130$ and $0.25 \leq d/\delta^* \leq 4$.

Stability setup to excite the boundary layer

- We linearize the flow around a steady base flow.
- We add a stochastic forcing term in the upstream part of the boundary layer as a form of white Gaussian noise.



Numerical predictions of Δn

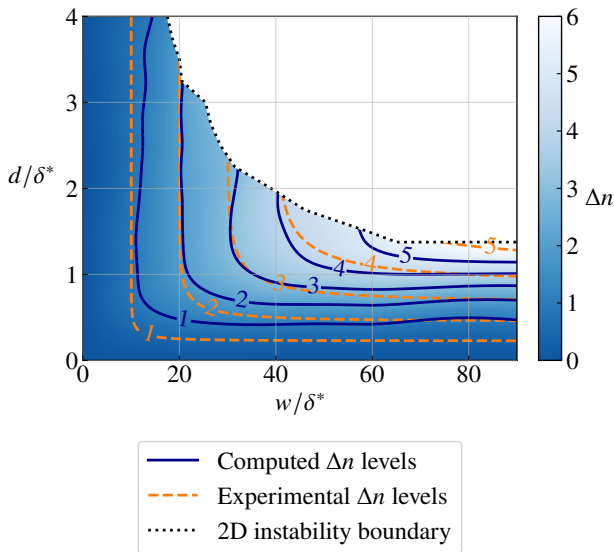


Figure: Contours of Δn in the w - d parameter space.

3d effects

- We perform single mode stability analysis to find the largest growth rate of 3D modes as a function of the spanwise wavenumber β .

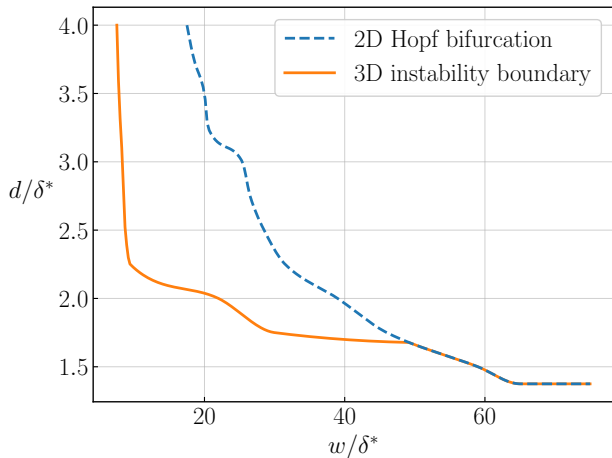
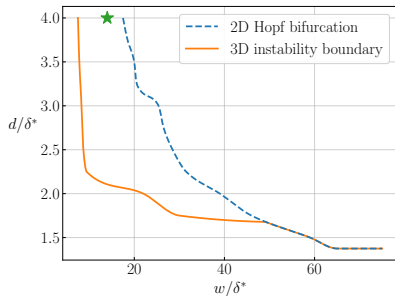


Figure: Bifurcation boundaries for the 2D and 3D instabilities.

3d non-linear results



- Long streaky structures appear in the downstream part of the gap
- Chaotic behaviour is observed in the gap, but insufficient to destabilise the system.

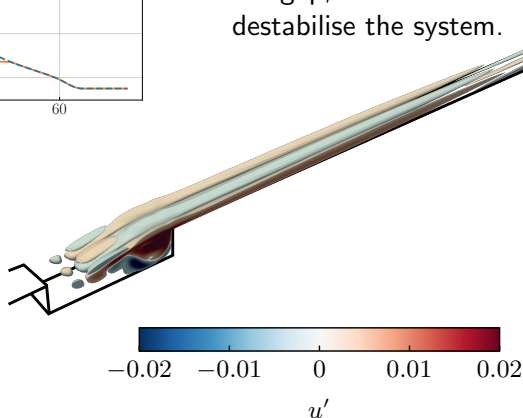
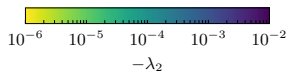
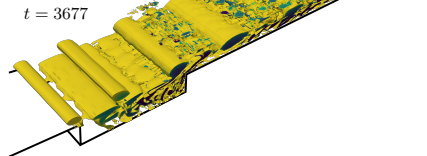
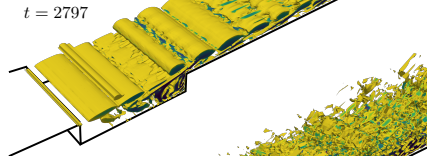
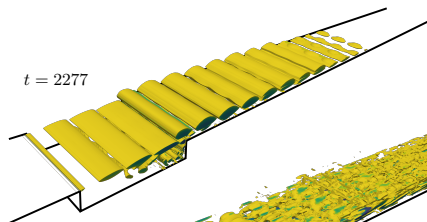
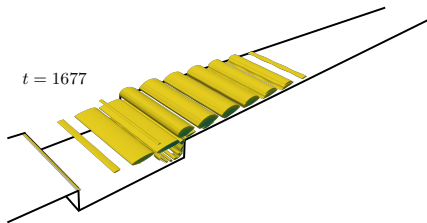
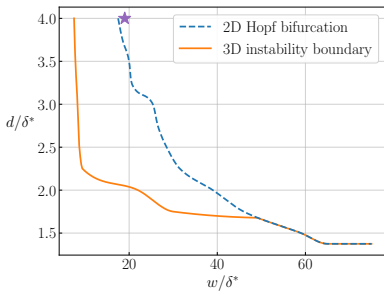


Figure: Most unstable mode for the 3D instability.

3d transition



Mach number effects

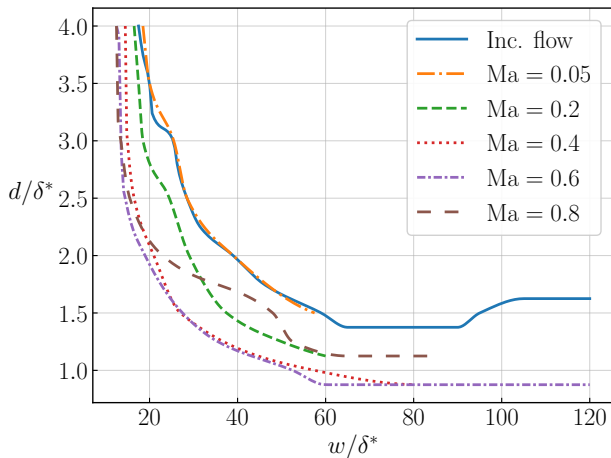


Figure: Hopf bifurcation boundaries for different Mach numbers.

Conclusions

- Very good agreement between the numerical predictions (2d inc.) and the experimental (3d low Mach wind tunnel) results.
- The 3D instability doesn't seem to be strong enough to destabilise the downstream boundary layer. Only after the 2D instability sets in, the transition location advances till the downstream edge of the gap.
- For deep gaps, Mach number has a destabilising effect, while for shallow gaps, higher Mach numbers lead to later transition.

Future work - compressible linear stability

- Extend the linear study to the compressible regime.
- We need a linear solver for the compressible Navier-Stokes equations.

Main aim:

- Adapt the Modified Arnoldi algorithm to the compressible Navier-Stokes equations.

A temporary solution: Jacobian free approach

Navier-Stokes equations can be written in the form

$$\partial_t \mathbf{q} = \mathbf{N}(\mathbf{q})$$

Assume we want to linearize $\mathbf{N}$ around a base flow $\bar{\mathbf{q}}$ so that we have:

$$\partial_t \mathbf{q}' = \mathbf{L}(\bar{\mathbf{q}}) \mathbf{q}' \quad \text{with} \quad \mathbf{L}(\bar{\mathbf{q}}) = \frac{\partial \mathbf{N}}{\partial \mathbf{q}}(\bar{\mathbf{q}})$$

We can write evolution operators for both the non-linear and linearized systems as

$$\mathbf{q}_t = \mathcal{A}_t^{\text{nonlin}} \mathbf{q}_0 \quad \text{and} \quad \mathbf{q}'_t = \mathcal{A}_t^{\text{lin}}(\bar{\mathbf{q}}) \mathbf{q}'_0$$

and we can relate them as

$$\mathcal{A}_t^{\text{lin}}(\bar{\mathbf{q}}) = \frac{\partial \mathcal{A}_t^{\text{nonlin}}}{\partial \mathbf{q}_0}(\bar{\mathbf{q}})$$

If we seek disturbances of the form

$$\mathbf{q}'(\mathbf{x}, t) = \hat{\mathbf{q}}(\mathbf{x})e^{\lambda t},$$

the linearized problem becomes

$$\mathbf{L}(\bar{\mathbf{q}})\hat{\mathbf{q}} = \lambda\hat{\mathbf{q}}.$$

We are usually interested in the eigenvalues of $\mathbf{L}(\bar{\mathbf{q}})$ with largest real part (Arnoldi algorithm), which are equivalent to the eigenvalues of the evolution operator

$$\mathcal{A}_t^{\text{lin}} = e^{\mathbf{L}(\bar{\mathbf{q}})t}$$

with largest magnitude (Modified Arnoldi algorithm).

Using the relation

$$\mathcal{A}_t^{\text{lin}}(\bar{\mathbf{q}}) = \frac{\partial \mathcal{A}_t^{\text{nonlin}}}{\partial \mathbf{q}_0}(\bar{\mathbf{q}})$$

we can compute the evolution of the linear system as a finite-difference approximation

$$\mathcal{A}_t^{\text{lin}}(\bar{\mathbf{q}})\mathbf{v} \approx \frac{\mathcal{A}_t^{\text{nonlin}}(\bar{\mathbf{q}} + \varepsilon \mathbf{v}) - \mathcal{A}_t^{\text{nonlin}}(\bar{\mathbf{q}})}{\varepsilon},$$

where $\varepsilon = \|\bar{\mathbf{q}}\|_2 \sqrt{\epsilon_{\text{machine}}}$ is chosen in a similar fashion to the one we use for the nonlinear solve of Compressible NS eqs.

Num. of computations:

- 1st instability: If $\bar{\mathbf{q}}$ is an “accurate” fixed point of $\mathcal{A}_t^{\text{nonlin}}(\bar{\mathbf{q}})$ then $\mathcal{A}_t^{\text{nonlin}}(\bar{\mathbf{q}}) \approx \bar{\mathbf{q}}$. $\implies N_{\text{iterArnoldi}}$ evaluations of $\mathcal{A}_t^{\text{nonlin}}$.
- 2nd instability: If $\bar{\mathbf{q}}$ is a limit cycle, then we need to evaluate $\mathcal{A}_t^{\text{nonlin}}(\bar{\mathbf{q}})$ once at the beginning of the algorithm and store the result for the rest of the iterations. $\implies N_{\text{iterArnoldi}} + 1$ evaluations of $\mathcal{A}_t^{\text{nonlin}}$.